
Invariance is not robustness: what predicts adversarial robustness, from CIFAR classifiers to CLIP encoders

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Abstract

A model that assigns the same label to a shifted view of an image is often called robust, and this shift-invariance is offered as an explanation for, or a route to, adversarial robustness. We show that this is the wrong lens, and identify what predicts robustness instead: a margin-to-sensitivity ratio measured in the attack’s own norm, and the shape of the gradient behind it. Invariance is about ignoring changes that should not matter; adversarial robustness is about not being flipped by a small worst-case change. These are different directions, and the invariance metric the field tracks, shift-consistency, is *saturated* on modern models: across sixteen frozen vision-language encoders it ranges only from 0.96 to 0.99, robust or not, so it has no room left to order them. A threat-matched margin-to-Lipschitz ratio η/L , the classification margin divided by the dual-norm size of its input gradient, does order robustness, from small CIFAR classifiers to a ResNet and, per image, within frozen CLIP encoders, and it is a certificate we complete to a two-sided bracket. The much-cited “invariance helps robustness” effect is a weak-attack artifact: it appears under the fast gradient sign method and vanishes under AutoAttack. We then close the one gap this diagnostic leaves. The ratio *detects* robustness but does not *rank* it among already-robust models; the missing ingredient is the *shape* of the gradient, its anisotropy $A = \|\nabla M\|_1/\|\nabla M\|_2$, which ranks thirty official RobustBench models at Spearman -0.79 , and -0.77 after controlling for clean accuracy. We identify the mechanism, an effective-dimension margin deficit, and separate it from the curvature account. Finally, the saturation is modality-general: on zero-shot CLIP both visual (shift) and textual (paraphrase) invariance saturate, and η/L predicts the image-adversarial axis while neither consistency metric does. Throughout we use AutoAttack as the arbiter and check for gradient masking.

1 Introduction

Call an image classifier *shift-invariant* if its prediction is unchanged by translations of the input. Two literatures disagree about what this buys. Ge et al. [10] show, for a binary problem in which each class is one image and all its shifts, that a fully shift-invariant network is fooled by an ℓ_2 perturbation of order $1/\sqrt{d}$ while a fully connected network needs order 1: invariance *reduces* robustness. A separate architectural line points the other way. Anti-aliased pooling raises shift-consistency, how often a shift leaves the prediction unchanged, along with clean accuracy and corruption robustness [31], and reducing aliasing has since been reported to raise *adversarial* robustness too [11, 23]: invariance *helps*. A third line proves a trade-off between geometric and adversarial robustness [15, 26].

These statements are not contradictory. They are regimes of one quantity, a margin-to-Lipschitz ratio. For an input x with label y , let the margin be $M(x) = f_y(x) - \max_{j \neq y} f_j(x)$, the gap between the correct logit and the strongest competitor. The per-image ratio is $M(x)/\|\nabla_x M(x)\|_q$, the margin divided by how fast it changes when the input is perturbed, measured in the norm dual to the attack

($q=1$ for an ℓ_∞ attack, $q=2$ for ℓ_2). A first-order argument gives a certificate: the adversarial radius of x is at least this ratio, because a perturbation of size δ moves the margin by at most $\|\nabla_x M\|_q \|\delta\|$. For a whole model we summarize with η/L , the mean margin η over the mean gradient size L on correctly classified clean images. We always match the norm to the attack and call this the *threat-matched* ratio. It has two properties that make it the right scalar. It needs no attack to compute. And it is *gauge-free*: rescaling the logits $f \mapsto cf$ multiplies η and L by the same c , so the ratio, and the input-space radius it lower-bounds, are unchanged. A model’s logit temperature cannot inflate or deflate it.

The reason invariance and robustness come apart is geometric. Invariance is being flat *along* the shift orbit, a direction that stays inside the correct class region. Robustness is the *distance* to the decision boundary, a different, near-orthogonal direction. A model can have a long flat orbit (very invariant) and still sit a hair from the boundary (fragile). We call this the *dissociation*, and it is the paper’s organizing observation: shift-invariance stays high and flat while adversarial robustness varies widely across models, so one cannot be read off the other. Formally, imposing invariance projects the classes onto the invariant subspace, and robustness is a distance to the boundary (Figure 1); measuring one says little about the other.

Contributions.

- **Invariance is saturated and does not rank robustness (§3).** Shift-consistency ranges only 0.96 to 0.99 across sixteen frozen encoders, robust or not; once clean accuracy is controlled it does not predict corruption robustness; and in a capacity-matched CNN dissection it does not order adversarial robustness and anti-predicts it under adversarial training.
- **The “invariance helps” effect is a weak-attack artifact (§4).** On every non-robust encoder, single-step FGSM reports 16 to 61% robust accuracy that AutoAttack erases to zero; retraining the two contrary operators under their own recipe reproduces their advantage under FGSM and removes it under a strong attack.
- **η/L predicts robustness, with theory (§5).** The threat-matched ratio orders robustness across three datasets, two threat models, a PreActResNet-18, and frozen CLIP encoders per image; it is a certificate $r \geq \eta/L$ that we complete to a two-sided bracket $\eta/L \leq r \leq \eta/\alpha$ with an explicit reachability slope α .
- **Detecting is not ranking: gradient shape and its mechanism (§6).** Among already-robust models η/L no longer orders robustness; the gradient’s anisotropy $A = \|\nabla M\|_1 / \|\nabla M\|_2$ does, at Spearman -0.77 over thirty RobustBench models after a clean-accuracy control. We identify the mechanism, an effective-dimension margin deficit $r \approx (\eta/L_1)/(1 + CA)$, show it is transverse rather than on-direction curvature, and place it as a relaxation of Simon-Gabriel et al. [24].
- **Robust training re-couples margin and sensitivity (§7).** The per-image relationship between margin and gradient size flips sign under adversarial training, a consistent fingerprint of what robust training changes.
- **Invariance saturates across modalities (§8).** On zero-shot CLIP, textual (paraphrase) invariance is as saturated as visual (shift) invariance, and η/L predicts the image-adversarial axis while neither consistency does, so the saturation of invariance carries over to text while the margin diagnostic stays image-specific and the wide adversarial axis stays on the image side.

2 Setup and definitions

We study classifiers $f : \mathbb{R}^d \rightarrow \mathbb{R}^K$ and the margin $M(x) = f_y(x) - \max_{j \neq y} f_j(x)$ at the correct label y , which is positive exactly when x is classified correctly. The *robust radius* $r(x)$ is the smallest $\|\delta\|$ with $M(x + \delta) \leq 0$; robust accuracy at budget ϵ is the fraction of test points with $r(x) > \epsilon$, which we always measure with AutoAttack [6] and cross-check against a per-sample minimum-norm attack for the radius. *Shift-consistency* is the fraction of translations of x that leave the top-1 label unchanged.

The threat-matched ratio and its shape. The per-image ratio is $M(x)/\|\nabla_x M(x)\|_q$ with q dual to the attack norm; the summary η/L uses the means over correctly classified clean images. We also

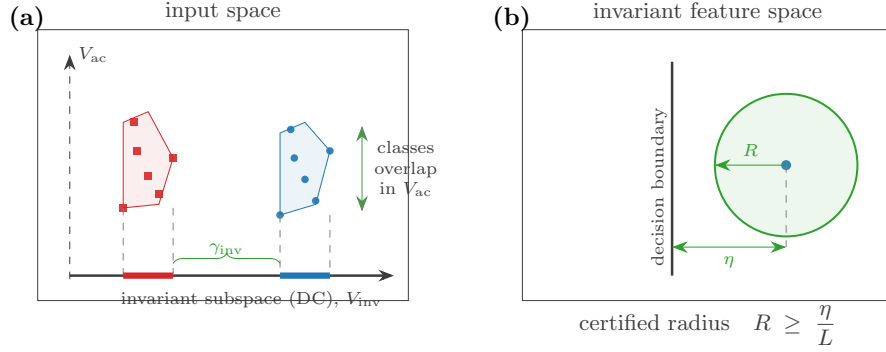


Figure 1: The two quantities the paper separates. **(a)** Imposing shift-invariance projects each class onto the invariant subspace V_{inv} (the DC direction); the classes can overlap in the anti-invariant part V_{ac} , and the surviving discriminative margin γ_{inv} is their separation along the invariant direction (Theorem 1). **(b)** Robustness is a distance: the certified radius satisfies $R \geq \eta/L$, the margin over the sensitivity.

read the *shape* of the gradient through its anisotropy

$$A(x) = \frac{\|\nabla_x M(x)\|_1}{\|\nabla_x M(x)\|_2} \in [1, \sqrt{d}],$$

which is near 1 when the gradient is concentrated on a few coordinates and near $\sqrt{d}$ when it is spread uniformly. Note $A = (\eta/L_2)/(\eta/L_1)$, so it is the same margin and gradient read as a ratio of the two threat-matched ratios; like η/L it is gauge-free.

Gauge-freeness as an organizing principle. Because $f \mapsto cf$ scales η and L together, robustness attaches to the ratio, not to the margin or the Lipschitz term alone; those two are individually gauge-dependent, so “larger margin” or “smaller Lipschitz” is well posed only once a scale is fixed. We anchor every robustness claim on the gauge-invariant quantities η/L , A , and the certified radius, and read any margin/Lipschitz split at the training-fixed logit scale.

Three model families. Our evidence spans three families, each answering a different question. *Small CIFAR, MNIST, and Fashion-MNIST classifiers and a PreActResNet-18*, where we build capacity-matched architectures so that only the shift-invariance mechanism differs, isolate the dissociation (§5). *Thirty adversarially trained CIFAR-10 models from RobustBench*, an independent family we did not train, test whether the gradient’s shape ranks robustness (§6). *Sixteen frozen vision-language (CLIP) image encoders* (ten of them adversarially robust) are where invariance is saturated (§3) and where a per-image analysis removes the adversarial-training confound (§5).

3 Invariance is saturated and does not rank robustness

The invariance metric the anti-aliasing literature tracks cannot separate modern models. Across sixteen frozen vision-language image encoders, robust and non-robust alike, shift-consistency ranges only from 0.963 to 0.988: every modern encoder is already near-perfectly shift-consistent, so the metric has no room to order them on any kind of robustness. Once clean accuracy is controlled it does not predict common-corruption robustness either. The same squeeze appears on strong CNN backbones.

This has a direct empirical consequence, the dissociation defined above. In a capacity-matched architectural dissection (four operators built to have identical parameter counts, so only the shift-invariance mechanism differs), shift-consistency does not order adversarial robustness, and under adversarial training it *anti-predicts* it: the exactly shift-invariant operator has perfect consistency and the *lowest* robust accuracy. The threat-matched η/L orders robustness instead. Table 1 previews the pattern that the next sections make precise across settings.

Table 1: The dissociation across settings. η/L orders robustness where shift-consistency does not. Values correlate with the robust radius (standard training) or threat-matched AutoAttack (adversarial training): Pearson for the CIFAR and ResNet grids, Spearman for the per-image CLIP result (within a fixed encoder).

setting	η/L vs robustness	shift-consistency vs robustness	reference
CIFAR-10 classifiers (standard)	+0.998	uninformative (-0.33 , n.s.)	§5
PreActResNet-18 (AT)	+0.91	anti-predicts (-0.81)	§5
Frozen CLIP encoders (per image)	+0.74 to +0.96	+0.37 to ≈ 0	§5

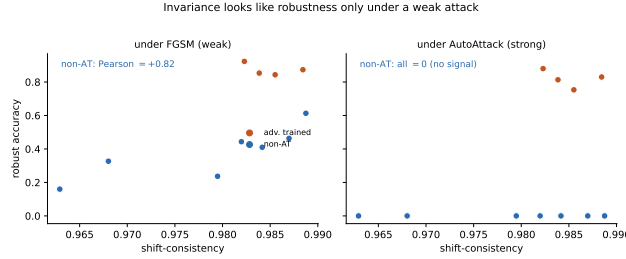


Figure 2: Same encoders and budget. Shift-consistency looks like it predicts robustness under FGSM (left) and predicts nothing under AutoAttack (right), because FGSM overstates the robustness of non-robust encoders that AutoAttack drives to zero.

4 The “invariance helps” effect is a weak-attack artifact

Reports that architectural invariance confers adversarial robustness without adversarial training are measured with weak attacks. On every non-robust encoder, single-step FGSM reports 16 to 61% robust accuracy, and AutoAttack erases it to zero (Figure 2). Under FGSM the “invariance helps” ordering appears, a real trend; under AutoAttack there is no robustness left to order and the trend disappears. This is the classic gradient-masking signature [1]: a weak attack overstates robustness, and the apparent correlation with invariance rides on that overstatement. We confirm the same on CNNs: retraining the two operators reported to make invariance help [23, 29] under their own code and recipe, their advantage reproduces under FGSM and vanishes under AutoAttack. We check for gradient masking throughout, and find none where we make a positive η/L claim: AutoAttack robust accuracy stays at or below matched multi-step PGD in every adversarially trained cell, and on the frozen encoders the parameter-free ensemble does not fall below its APGD component.

5 The margin-to-Lipschitz ratio predicts robustness

Theory: a two-sided bracket. The certificate $r \geq \eta/L$ is the standard first-order lower bound. We complete it to a two-sided bracket. Let α be a reachability slope: the slowest rate at which the margin can fall as one walks from x toward the boundary (a lower bound on that rate). Then

$$\eta/L \leq r \leq \eta/\alpha, \quad \text{so} \quad 1 \leq \frac{r}{\eta/L} \leq \kappa := \frac{L}{\alpha}.$$

The gap is a local condition number $\kappa = L/\alpha$; when κ is close to one, η/L is not just a bound but tight, essentially the radius itself. For linear features the two sides coincide ($\kappa=1$, $r = \eta/\|w\|$, recovering Hein and Andriushchenko [12]); we verify numerically that the true radius lies strictly inside the bracket with $\kappa \in [1.6, 2.4]$. This completes the missing upper arm of the margin-Lipschitz bound of Tsuzuku et al. [27]. The gauge-freeness and the structural theory behind η (the invariant linear margin is the separation of the data projected onto the invariant subspace, which recovers the $1/\sqrt{d}$ collapse of Ge et al. [10]) are in Appendix A.

Evidence across families. The same law holds from small CNNs to frozen foundation models.

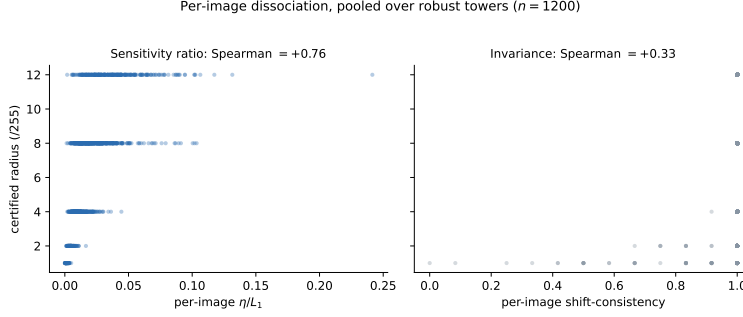


Figure 3: Per image, pooled over the robust CLIP encoders ($n=1200$; each comparison is within one encoder, so adversarial training cannot confound it). The threat-matched ratio η/L_1 (left, Spearman $+0.76$) orders each image’s certified radius; shift-consistency (right, $+0.33$) does not. Per-encoder ranges ($+0.74$ to $+0.96$ versus $+0.37$ to near zero) are in the text.

- **CIFAR-10, standard training.** Across a dense 24-cell capacity-matched grid, η/L orders the ℓ_2 robust radius at Pearson $+0.998$ (bootstrap CI $[+0.997, +0.999]$, $p < 10^{-3}$); shift-consistency does not (Pearson -0.33 , CI $[-0.68, +0.09]$, $p = 0.11$).
- **PreActResNet-18, adversarial training.** Over twelve architecture-by-width cells the threat-matched $\eta/\|\nabla M\|_1$ orders AutoAttack robustness at Pearson $+0.91$ ($p < 10^{-3}$), carrying robustness information beyond clean accuracy (partial correlation $+0.89$ controlling for clean accuracy); shift-consistency anti-predicts at -0.81 (partial -0.82). At ImageNet-100 resolution, where clean accuracy is essentially constant across arms (0.641 to 0.649), the same threat-matched law holds at $+0.90$ while shift-consistency anti-predicts at -0.87 , so the prediction is not a restatement of clean accuracy.
- **Frozen CLIP encoders, per image.** Within each of ten adversarially robust encoders, where adversarial training cannot act as a confound, the per-image η/L orders the per-image adversarial radius at Spearman $+0.74$ to $+0.96$, while per-image shift-consistency orders it from $+0.37$ down to essentially zero, and the gap is significant in a paired test in every encoder (Figure 3).

Fitted without a held-out learnable-invariance operator, η/L predicts that operator’s robustness out of sample, and the same threat-matched law holds when we intervene on the training *data* instead of the architecture, through diffusion-augmented adversarial training (Appendix B).

6 Detecting is not ranking: gradient shape and its mechanism

The ratio η/L cleanly *separates* robust models from fragile ones, but *across* the already-robust models it no longer orders them: two encoders with the same η/L can differ in robustness, and η/L_1 ranks the thirty robust RobustBench models only weakly (Figure 4b). This is a different axis from the per-image ranking of §5, which holds *within* a single encoder; here we compare *across* robust models. The ratio uses the *size* of the gradient; ranking robust models needs its *shape*.

Anisotropy ranks robust models. An ℓ_∞ attack pushes on every coordinate at once. If the margin’s gradient is concentrated on a few coordinates, most of the budget is wasted and the model holds; if it is spread over many, the attack engages all of them. The anisotropy A measures exactly this spread. Across thirty official RobustBench CIFAR-10 ℓ_∞ models (a family independent of the CLIP encoders), lower A means more robust: Spearman(A , robust) = -0.79 over the thirty models (bootstrap interval $[-0.92, -0.56]$), which survives controlling for clean accuracy (partial -0.77) and for η/L_1 itself (partial -0.65). Restricted to the twenty-nine robust models, dropping the single non-robust one, the ranking is -0.77 (Figure 4a). The size η/L_1 detects robustness but ranks it only weakly (Figure 4b). The same negative ranking appears within the robust CLIP encoders.

The mechanism: an effective-dimension margin deficit. Why does spread hurt? The first-order estimate η/L_1 is only the leading term. A real attack also harvests the gradient’s variation across coordinates, and it collects more of that variation when the gradient is spread over more coordinates.

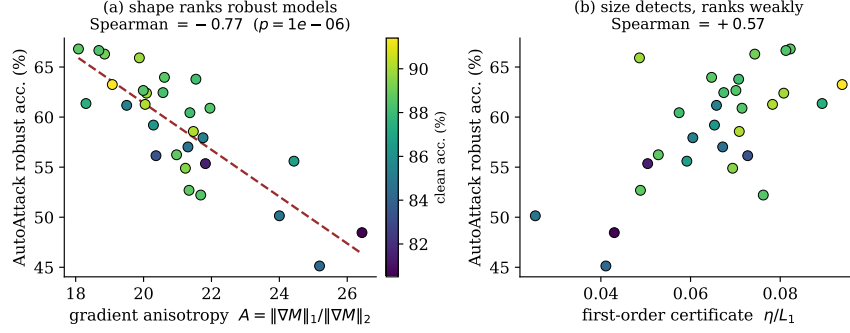


Figure 4: Thirty RobustBench CIFAR-10 models. **(a)** the gradient’s shape A ranks robustness (Spearman -0.77 , surviving a clean-accuracy control); **(b)** the size η/L_1 detects robustness but ranks it only weakly.

This shrinks the true radius below η/L_1 by a factor that grows with the spread:

$$r \approx \frac{\eta/L_1}{1 + C A}, \quad (1)$$

with a model-level constant C . In a controlled model with the margin and the total gradient energy held fixed and only the effective dimension varied, the true multi-step ℓ_∞ radius orders perfectly against A (Spearman -1.0), and the deficit scales as $\sqrt{\text{active dims}} = A$ (log-log slope 0.53). The effect is *transverse*, spread over the many coordinates, not the curvature straight along the attack: we measured the on-direction curvature and it does *not* rank robustness, and A does not track it, exactly as in the real encoders. So the curvature-regularization account [21] is not the ranking mechanism here.

Relation to prior work, and scope. The magnitude-versus-spread reading is the relaxation of Simon-Gabriel et al. [24], whose dimension law assumes equal gradient variance per coordinate and so fixes $A \propto \sqrt{d}$; we let A vary at fixed d and find it tracks robustness. The nearest metric is the participation ratio of the input gradient [19], used there to *detect* catastrophic overfitting during training rather than to rank models across a family. The synthetic model in Eq. 1 is a self-consistent stand-alone that assumes per-coordinate independence, not a derivation from the network; the strong signal is between models, so C is a model-level property. We therefore present the deficit as a proposed mechanism, consistent with every control and refuting the curvature alternative, rather than as a first-principles theorem. The law also generalizes to two low-dimensional families. On MNIST and Fashion-MNIST a fixed-budget grid compresses A into a narrow band and it does not rank, but widening the range with an adversarial-training-strength sweep restores it, so the compression, not the low input dimension, was the obstacle. Over twenty-four MNIST cells (with A now spanning $[2.4, 16.6]$) A ranks threat-matched ℓ_∞ robustness at the informative budgets, -0.83 ($\varepsilon=0.2$) and -0.89 ($\varepsilon=0.3$) on MNIST (partial controlling clean accuracy -0.75 and -0.74), and -0.61 ($\varepsilon=0.1$) to -0.80 ($\varepsilon=0.15$) over twenty-four Fashion-MNIST cells (partial -0.53 and -0.67). Two caveats carry over: the sign is correct only in the training norm (in the mismatched ℓ_2 radius on MNIST A takes the wrong sign, $+0.48$ raw, collapsing to $+0.10$ once clean accuracy is controlled), and on this single-knob sweep A is collinear with training strength (Spearman(A, ε) ≈ -0.9), so the cleaner cross-model evidence remains the diverse RobustBench panel, where A ranks robustness even after controlling for η/L_1 (Appendix B).

7 Robust training re-couples margin and sensitivity

The two ingredients of η/L are coupled, and adversarial training changes how. In a standard model the per-image margin and gradient size are anti-aligned: high-margin images sit in flatter regions with *smaller* gradients (within-model Spearman $\rho(M, \|\nabla M\|_2) \approx -0.5$ in every one of the twelve standard models), so margin and robustness reinforce. Adversarial training flips this to positive ($\approx +0.5$ in every one of the twenty-four robust models): a robust model buys extra margin partly by raising local sensitivity, so the margin-to-sensitivity ratio is self-limiting (Figure 5). The per-image

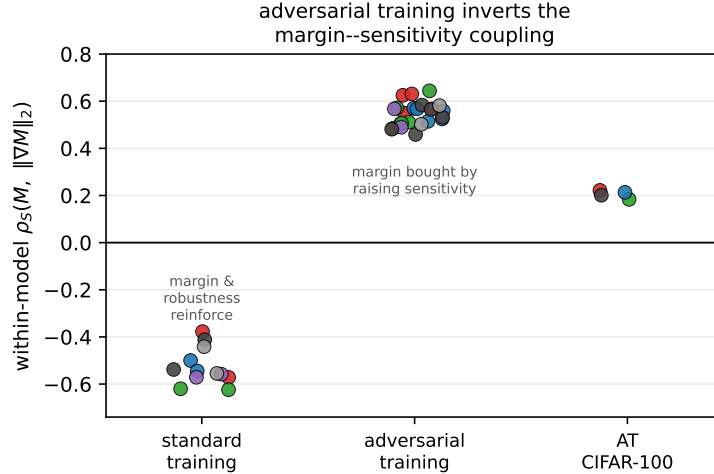


Figure 5: The within-model coupling between margin M and input-gradient norm $\|\nabla M\|_2$ flips sign under adversarial training: negative under standard training (high-margin points are flatter) and positive after adversarial training (margin is bought by raising sensitivity), for every model, across the ResNet grid on CIFAR-10 and CIFAR-100.

coupling flips sign, consistently across dozens of models. This is why one ratio predicts robustness in both regimes: the margin and the gradient size, read separately, are anti-aligned in standard models and aligned after robust training, and only their ratio η/L tracks robustness through both. The same re-coupling explains why η/L is a diagnostic and not a trainable target: because the ratio is scale-invariant its direct maximizer is the constant classifier (Lemma 3), and first-order attempts to raise it collapse the margin (Appendix B).

8 Invariance saturates across modalities

The saturation of §3 is not specific to pixel shifts. On zero-shot CLIP over ImageNet-100 we lay out a two-by-two: two modalities (image, text) crossed with two perturbation types over a semantically-null family, an average-case invariance and a worst-case adversarial version. Images are shifted (average) or attacked with full targeted AutoAttack (worst case); prompts are reworded into eleven curated meaning-preserving paraphrases (average, paraphrase-consistency PC) or scored by the worst of them (worst case, adversarial-paraphrase accuracy S_{text}). The panel is seventeen encoders: four non-robust, thirteen adversarially robust, including two Sim-CLIP towers and one text-hardened tower (Figure 6).

Both invariance cells saturate: shift-consistency is 0.977 ± 0.009 and paraphrase-consistency 0.977 ± 0.007 . Only the image-adversarial cell carries cross-encoder spread ($S = 0.385 \pm 0.244$, range $[0, 0.69]$, twenty-eight times the shift-consistency spread), and the threat-matched η/L_1 predicts it ($+0.80$, CI $[0.44, 0.95]$; partial controlling clean accuracy $+0.82$; per image within the robust encoders $+0.89$, $n=2600$), while the consistencies do not (shift $+0.21$, paraphrase -0.30 , both intervals spanning zero). The one place the clean two-by-two does not materialize is the text-adversarial cell: the worst of eleven curated paraphrases is itself a mild perturbation, so S_{text} stays narrow (0.896 ± 0.029 , four times the paraphrase spread), and η/L_1 , an image-side quantity, is orthogonal to its residual variation once clean accuracy is controlled (-0.54 raw, partial $+0.01$, so the raw anti-correlation is a clean-accuracy artifact). Paraphrase-consistency co-moves with S_{text} ($+0.98$), but that is average against worst case over the same narrow family. A single text-hardened encoder, adversarially fine-tuned on the text tower, has the highest S_{text} and paraphrase-consistency in the panel despite a middling vision η/L_1 , which places text robustness as a separate axis raised by text training and not by the vision margin.

Two things follow. The saturation of invariance metrics is modality-general: textual paraphrase invariance is as saturated as visual shift invariance, and no more predictive of adversarial robustness. And the margin diagnostic is modality-specific: η/L_1 read on the image tower governs the image-

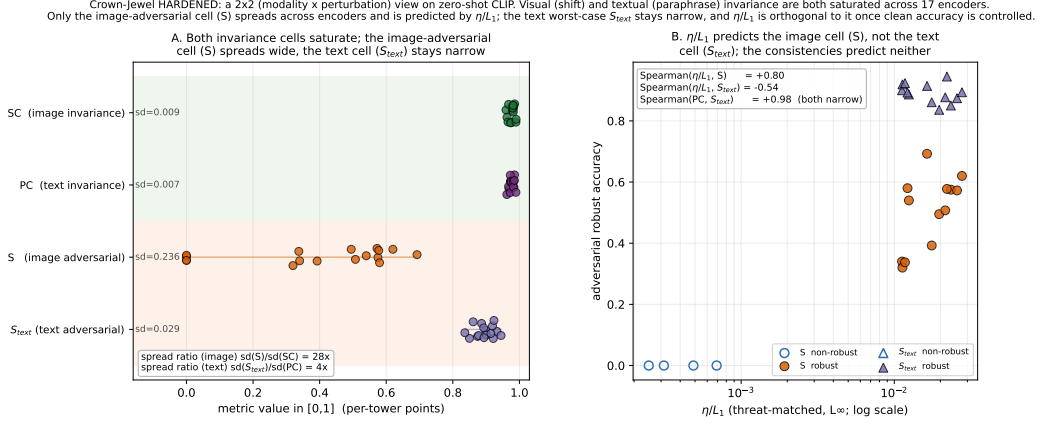


Figure 6: Zero-shot CLIP, ImageNet-100, seventeen encoders, full targeted AutoAttack. **(a)** both invariance cells (shift-consistency, paraphrase-consistency) saturate near 1; the image-adversarial cell S spreads widely while the text-adversarial cell S_{text} (worst of eleven paraphrases) stays narrow. **(b)** η/L_1 orders the image-adversarial cell (+0.80) but not the text cell (−0.54 raw, and zero after controlling for clean accuracy), and neither consistency orders either. The continuous ℓ_∞ image attack and the discrete paraphrase set are both worst cases over a semantically-null family; the small paraphrase set is the milder perturbation.

adversarial axis, while a wide text-adversarial axis, if one is wanted, needs a stronger textual perturbation than a small curated paraphrase set supplies.

9 Related work

Invariance and robustness. Ge et al. [10] give the trade-off on single-orbit data; Kamath et al. [15], Tramèr and Boneh [26] prove tensions between geometric and adversarial robustness; the anti-aliasing line [31, 11, 23] and the equivariance line [29] report the opposite sign. These locate as positions on the threat-matched η/L axis: Ge’s collapse and the trade-off results are the small-margin corner, where invariance costs margin and η/L falls, while the “helps” reports are a weak-attack reading that does not survive AutoAttack rather than a third regime. **Attack-free robustness estimates.** CLEVER [30], the generative GREAT Score [17], and margin-Lipschitz certificates [12, 27] estimate a radius; we make the estimate gauge-free, threat-matched, and two-sided, and add the anisotropy refinement for ranking robust models. Certified defenses such as randomized smoothing [5] also return a radius, but through a smoothed classifier and sampling; η/L is complementary, a first-order read of the base model that needs no smoothing and no attack. **Gradient geometry.** Simon-Gabriel et al. [24] tie vulnerability to gradient norm and dimension under equal per-coordinate variance; we relax that assumption. Moosavi-Dezfooli et al. [21] regularize curvature for robustness; we show on-direction curvature does not rank robust models and the transverse effective-dimension deficit does. The gradient participation ratio [19] is used for overfitting detection, not cross-model ranking.

10 Limitations and discussion

The tower-level CLIP correlation rests on a small number of robust encoders, so the per-image result is the load-bearing one; the hardened multimodal benchmark (§8) expands the panel and uses full AutoAttack. The deficit law of Eq. 1 is a proposed, well-controlled mechanism rather than a network-level theorem, and the ranking it drives is range-dependent: it needs a wide robustness axis, which on low-dimensional families appears only once a training-strength sweep opens it. Finally η/L is a diagnostic to read off a trained model, not a target to train toward. Within those limits, one idea, the margin relative to the sensitivity, unifies a split literature, corrects a common evaluation mistake, and turns a vague intuition about invariance into something measurable: invariance is not

robustness, the margin relative to the sensitivity is, its shape is what separates the strong models, and the saturation of invariance carries from pixel shifts to text.

Reproducibility and ethics. All quantities are computed with released attack code (AutoAttack, a minimum-norm attack) on standard benchmarks; the contribution is diagnostic and defensive, introducing no new attack. Code, configurations, and per-sample outputs are released.

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A Full structural and optimization theory

This appendix gives, in full, the proofs the main text states or sketches, and the supporting structural results. It is self-contained: §A.1 fixes notation, §A.2–§A.7 prove the six load-bearing results in the order announced in the main body. Where the main text scopes a claim as proposed or conditional (the effective-dimension deficit of §6, the rich-regime KKT conjecture of the optimization discussion), that scoping is preserved verbatim here.

A.1 Setup and standing notation

We work with one-dimensional signals $x \in \mathbb{R}^d$; the two-dimensional case replaces $\mathbb{Z}_d$ by $\mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2}$ and the cyclic discrete Fourier transform (DFT) by its two-dimensional analogue, without change to any statement. The cyclic group $G = \mathbb{Z}_d$ acts by the circular shift $(S_s x)_i = x_{(i-s) \bmod d}$; each S_s is an orthogonal permutation. The orbit of x is $\mathcal{O}(x) = \{S_s x\}_{s \in G}$, and a class set is *orbit-closed* if it is closed under all shifts. Let $V_{\text{inv}} = \{v : S_s v = v \ \forall s\}$ and $V_{\text{ac}} = V_{\text{inv}}^\perp$. The orthogonal projector onto V_{inv} is $\Pi_{\text{inv}} = \frac{1}{d} \mathbf{1} \mathbf{1}^\top = ee^\top$ with $e = d^{-1/2} \mathbf{1}$, so $V_{\text{inv}} = \text{span}(\mathbf{1})$ is the constant (DC, zero-frequency) line and $V_{\text{ac}} = \{v : \sum_i v_i = 0\}$ is its zero-mean (AC) complement. Write $f_{\text{dc}}(x) = e^\top x = d^{-1/2} \sum_i x_i$ for the unit DC functional. With the unitary DFT $\hat{x}_k = d^{-1/2} \sum_j x_j e^{-2\pi i j k / d}$ and $\omega = e^{2\pi i / d}$, a shift multiplies coefficient k by the unit phase ω^{sk} , so $|\hat{x}_k|^2$ is shift-invariant.

For a score f correct on (x, y) , the ℓ_p robust radius is $r_p(f, x) = \inf\{\|\delta\|_p : \text{sign} f(x + \delta) \neq y\}$; for a linear score $w^\top x + b$ with $yf(x) > 0$ this equals $y(w^\top x + b)/\|w\|_q$ (q dual to p). The margin at the correct label is $M(x) = f_y(x) - \max_{j \neq y} f_j(x)$, positive exactly when x is correct; $\eta = \mathbb{E}[M]$ and $L = \mathbb{E}\|\nabla M\|_q$ are the dataset means on correctly classified clean points. For finite sets A, B write the hard-margin separation $\text{Sep}(A, B) := \frac{1}{2} \text{dist}(\text{conv } A, \text{conv } B)$, zero when the hulls intersect. All statements match the combined main text’s notation and environments.

Lemma 1 (Fixed subspace for cyclic shifts). *For the cyclic shift action on $\mathbb{R}^d$, $V_{\text{inv}} = \text{span}(\mathbf{1})$, $\Pi_{\text{inv}} = d^{-1} \mathbf{1} \mathbf{1}^\top = ee^\top$, and $V_{\text{ac}} = \{v : \mathbf{1}^\top v = 0\}$. The shifts are simultaneously diagonalized by the DFT, $F S_s F^* = \text{diag}(\omega^{sk})_{k=0}^{d-1}$, each eigenvalue of modulus one, so $|\widehat{S_s x}_k|^2 = |\hat{x}_k|^2$ for every k .*

Proof. If $v \in V_{\text{inv}}$ then $S_1 v = v$ reads $v_i = v_{i-1}$ modulo d , so all coordinates are equal and $v \in \text{span}(\mathbf{1})$; conversely every constant vector is fixed by every S_s . The orthogonal projector onto $\text{span}(\mathbf{1})$ is $\mathbf{1} \mathbf{1}^\top / \|\mathbf{1}\|_2^2 = d^{-1} \mathbf{1} \mathbf{1}^\top = ee^\top$, with complement $\{v : \mathbf{1}^\top v = 0\}$. Every S_s is a circulant permutation, hence diagonalized by the DFT; its eigenvalues have modulus one, so shifting changes only Fourier phases and preserves each $|\hat{x}_k|^2$. $\square$

A.2 Result 1: the invariant linear margin equals projection separation

We restate Theorem 1 of the main text identically, prove it in full (reduction to one-dimensional DC thresholding; max-margin equals half the inter-interval distance; the $\pm \mathbf{1}/\sqrt{d}$ direction; recovery of the Ge $1/\sqrt{d}$ collapse), and give the finite-group generalization.

Theorem 1 (Invariant linear margin equals projection separation; restated). *A linear score $g(x) = w^\top x + b$ is shift-invariant for all x if and only if $w = c\mathbf{1}$. Consequently, for any finite separable classes X_+, X_- , the invariant ℓ_2 max-margin equals the separation of the DC-projected data,*

$$\gamma_{\text{inv}} = \frac{1}{2} \text{dist}(I_+, I_-) = \frac{1}{2} \text{dist}(\text{conv } \Pi_{\text{inv}} X_+, \text{conv } \Pi_{\text{inv}} X_-),$$

where $I_\pm = [\min_{x \in X_\pm} f_{\text{dc}}(x), \max_{x \in X_\pm} f_{\text{dc}}(x)]$, with maximizing direction $\pm \mathbf{1}/\sqrt{d}$ oriented toward the positive class. This holds for any finite separable dataset and does not require orbit-closure.

Proof. Characterization of invariant normals. Invariance of g under every shift means $w^\top S_s x = w^\top x$ for all x, s , i.e. $(S_{-s} w - w)^\top x = 0$ for all x , hence $S_{-s} w = w$ for all s . The only vectors fixed by every cyclic shift are the constants (Lemma 1), so $w \in V_{\text{inv}} = \text{span}(\mathbf{1})$, i.e. $w = c\mathbf{1}$; conversely every $c\mathbf{1}$ gives an invariant score.

Reduction to one-dimensional DC thresholding. With $w = c\mathbf{1}$ the score depends on x only through $f_{\text{dc}}(x) = \mathbf{1}^\top x / \sqrt{d}$, because $w^\top x = c\mathbf{1}^\top x = c\sqrt{d} f_{\text{dc}}(x)$. Thus the data reduce to their scalar DC values and the classes to the intervals $I_\pm = [\min_{X_\pm} f_{\text{dc}}, \max_{X_\pm} f_{\text{dc}}]$ on the line $\text{span}(e)$.

Max-margin on the line. The maximum-margin separator of two point sets on a line places the threshold midway in the gap between their convex hulls (here the two intervals $I_\pm$); its geometric margin is half the distance between the intervals, $\gamma_{\text{inv}} = \frac{1}{2} \text{dist}(I_+, I_-)$. Concretely, if I_+ lies to the right of I_- the optimal unit normal is $+e = +\mathbf{1}/\sqrt{d}$ and $\gamma_{\text{inv}} = \frac{1}{2}(\min_{X_+} f_{\text{dc}} - \max_{X_-} f_{\text{dc}})$; if I_- lies to the right the normal is $-e$; and when the intervals overlap the separation, hence the invariant margin, is 0. Identifying the projected line with $\text{span}(\mathbf{1})$ isometrically gives the convex-hull form $\frac{1}{2} \text{dist}(\text{conv } \Pi_{\text{inv}} X_+, \text{conv } \Pi_{\text{inv}} X_-)$. $\square$

Corollary 1 (Recovery of the Ge $1/\sqrt{d}$ collapse). *For the single white/black dot $x_+ = e_j$, $x_- = -e_j$ one has $f_{\text{dc}}(\pm e_j) = \pm 1/\sqrt{d}$, so Theorem 1 gives $\gamma_{\text{inv}} = 1/\sqrt{d}$ (half-gap convention; the full DC gap is $2/\sqrt{d}$), the collapse of Ge et al. [10]. The unconstrained max-margin of the two unshifted points $\{e_j\}, \{-e_j\}$ is 1, but on the orbit-closed dataset $\mathcal{O}(e_j)$ versus $\mathcal{O}(-e_j)$ the unconstrained linear max-margin is itself $1/\sqrt{d}$, so the 1-versus- $1/\sqrt{d}$ contrast is a property of the two-image protocol, not of invariance alone.*

Proof. $f_{\text{dc}}(\pm e_j) = e^\top(\pm e_j) = \pm 1/\sqrt{d}$ gives the invariant margin directly. On $\mathcal{O}(e_j)$ vs $\mathcal{O}(-e_j)$ every positive basis vector must be classified positive and every negative one negative by a single hyperplane; the symmetric max-margin solution is $w \propto \mathbf{1}$ with margin $1/\sqrt{d}$, so the unconstrained and invariant margins coincide there. $\square$

Proposition 1 (General finite-group invariant margin). *Let a finite group G act orthogonally on $\mathbb{R}^d$ by matrices $\{R_g\}$, with fixed subspace $V^G = \{v : R_g v = v \ \forall g\}$ and group-average projector $P_G = \frac{1}{|G|} \sum_g R_g$. A linear score $w^\top x + b$ is exactly G -invariant if and only if $w \in V^G$, and the maximum invariant ℓ_2 margin for finite separable classes is $\gamma_{\text{inv}} = \frac{1}{2} \text{dist}(\text{conv } P_G X_+, \text{conv } P_G X_-)$. Theorem 1 is the cyclic case $P_G = \Pi_{\text{inv}}$.*

Proof. P_G is the orthogonal projector onto V^G : group closure gives $P_G^2 = |G|^{-2} \sum_{g,h} R_g R_h = |G|^{-1} \sum_k R_k = P_G$; each R_g orthogonal gives $R_g^\top = R_{g^{-1}}$ and $P_G^\top = P_G$; its range is exactly V^G . Invariance of $w^\top x$ for all x is equivalent to $R_g^\top w = w$ for all g , hence (as G is a group) to $R_g w = w$, i.e. $w \in V^G$. For $w \in V^G$ we have $w = P_G w$, so $w^\top x = w^\top P_G x$: every invariant linear classifier on $X_\pm$ is an ordinary linear classifier on the projected sets $P_G X_\pm$, with the same normal and signed margins. The maximum unit-normal margin separating two finite sets A, B equals $\frac{1}{2} \text{dist}(\text{conv } A, \text{conv } B)$, attained by the unit vector along the closest-points segment of the hulls: if every point of A, B is at signed distance $\geq \gamma$ from a unit-normal hyperplane, so is every convex combination, so the hulls are at distance $\geq 2\gamma$; the segment midpoint realizes it. Apply this to $A = P_G X_+$, $B = P_G X_-$. $\square$

Corollary 2 (Projection cannot increase the linear max-margin). *With $\gamma_{\text{full}} = \frac{1}{2} \text{dist}(\text{conv } X_+, \text{conv } X_-)$ and γ_{inv} as above, $\gamma_{\text{inv}} \leq \gamma_{\text{full}}$, because orthogonal projectors are non-expansive: $\|P_G u - P_G v\|_2 \leq \|u - v\|_2$, so taking the infimum over the hulls preserves the inequality.*

A.3 Result 2: the two-sided bracket (a conditional converse)

We restate Proposition 2 of the main text and give the full proof, with the explicit reaching-path and co-Lipschitz(α) hypotheses, the exact linear case $\kappa = 1$ [12], the t^{2n+1} large- κ counterexample, and the note that it completes the missing upper arm of Tsuzuku et al. [27].

Proposition 2 (Two-sided bracket; restated). *Let g be continuous and L -Lipschitz in ℓ_2 on a convex set S containing x , its nearest boundary point, and the path γ below, correct at x with margin $\eta = y g(x) > 0$, and write $\mathcal{B} = \{z : g(z) = 0\}$. (Lower arm) $r_2(g, x) \geq \eta/L$. (Upper arm) if there is a continuous path $\gamma : [0, T] \rightarrow S$ with $\gamma(0) = x$, $g(\gamma(T)) = 0$, and a co-Lipschitz (lower-Lipschitz) bound $|g(\gamma(t)) - g(x)| \geq \alpha \|\gamma(t) - x\|_2$ up to the first boundary hit, then $r_2(g, x) \leq \eta/\alpha$,*

and necessarily $\alpha \leq L$. Hence

$$\frac{\eta}{L} \leq r_2(g, x) \leq \frac{\eta}{\alpha}, \quad \kappa := \frac{L}{\alpha} \geq 1,$$

so η/L determines the robust radius up to the local bi-Lipschitz condition number κ ; the certificate is exact ($r_2 = \eta/L$) iff $\kappa = 1$.

Proof. Convexity of S keeps the relevant segments inside S , so the local Lipschitz and co-Lipschitz constants apply along them.

Lower arm. For any δ with $\|\delta\|_2 < \eta/L$ the Lipschitz bound gives $|g(x + \delta) - g(x)| \leq L\|\delta\|_2 < \eta = |g(x)|$, so $g(x + \delta)$ keeps the sign of $g(x)$ and no perturbation below η/L flips the label; hence $r_2 \geq \eta/L$. This is the standard Lipschitz–margin certificate [12, 27].

Upper arm. At $t = T$, $g(\gamma(T)) = 0$ and $|g(x)| = \eta$, so $\eta = |g(x) - g(\gamma(T))| \geq \alpha\|\gamma(T) - x\|_2$, giving $\|\gamma(T) - x\|_2 \leq \eta/\alpha$. Since $\gamma(T) \in \mathcal{B}$ has $\text{sign}g(\gamma(T)) \neq y$, and by continuity points immediately beyond $\gamma(T)$ have strictly opposite sign, the perturbation $\delta = \gamma(T) - x$ is label-flipping and $r_2 \leq \|\gamma(T) - x\|_2 \leq \eta/\alpha$.

$\alpha \leq L$. As $g(\gamma(T)) = 0 \neq g(x)$ we have $\gamma(T) \neq x$, and the same secant obeys $\alpha\|\gamma(T) - x\|_2 \leq |g(\gamma(T)) - g(x)| \leq L\|\gamma(T) - x\|_2$, so $\alpha \leq L$ and $\kappa \geq 1$. Combining the arms gives the bracket, tight iff $\kappa = 1$. The upper arm is informative only when α is certified independently of r_2 : taking α as the secant slope of the minimal path itself makes $r_2 = \eta/\alpha$ a tautology, whereas the linear case and the power-spectrum surrogate below fix α from the feature structure. $\square$

Corollary 3 (Linear invariant features: the converse is exact). *If $g(x) = w^\top x + b$ with $w \in V_{\text{inv}}$ (so g reads x only through f_{dc}), then along the straight path $\gamma(t) = x - tw/\|w\|_2$ one has $|g(z) - g(x)| = |w^\top(z - x)| \leq \|w\|_2\|z - x\|_2$ with equality along $\pm w$, so $L = \alpha = \|w\|_2$, $\kappa = 1$, and the boundary is reached at distance $\eta/\|w\|_2$: the exact point-to-hyperplane radius $r_2 = \eta/\|w\|_2$ of Hein and Andriushchenko [12].*

Remark 1 (The large- κ counterexample: t^{2n+1}). There is no universal converse. For the shift-invariant scalar feature $\Psi_n(x) = t^{2n+1}$ with $t = f_{\text{dc}}(x)$, and points $x_+ = e$, $x_- = -e$, the signed feature margin is $\eta = 1$. On the segment $\{te : t \in [-1, 1]\}$ the Lipschitz constant is $L_n = \max_{|t| \leq 1} (2n+1)t^{2n} = 2n+1$, so the certificate gives $\eta/L_n = 1/(2n+1)$. But the boundary is $t = 0$, reached from e at Euclidean distance 1 (perturbations in V_{ac} do not change t), so $r_2 = 1$ and $r_2/(\eta/L_n) = 2n+1 \rightarrow \infty$. In the bracket language this is the large- κ end: the straight DC path has secant slope $\alpha = 1$, the upper Lipschitz is $L = 2n+1$, so $\kappa = 2n+1$ and the bracket $[1/(2n+1), 1]$ correctly contains $r_2 = 1$ at its η/α end. The failure of a universal converse is exactly the unboundedness of κ over the model class, not the absence of a bracket for a fixed model.

Remark 2 (Completing the Tsuzuku chain). Tsuzuku et al. [27] (their Eq. 5) bound r_2 only from below through successively tighter Lipschitz constants and state the top inequality (certificate $\leq$ true radius) cannot be guaranteed, measuring it empirically as a 1.8–2.4 $\times$ gap. The upper arm $r_2 \leq \eta/\alpha$ of Proposition 2 is the matching arm that turns their one-sided certificate into a two-sided bracket. It is deterministic and tied to a single fixed classifier, its looseness controlled by the explicit κ rather than, as in the randomized-smoothing converse of Cohen et al. [5], by a worst case over classifiers. We verify numerically that the true radius lies strictly inside the bracket with $\kappa \in [1.6, 2.4]$ for the power-spectrum surrogate.

For completeness we record the certifiable upper arm for the quadratic-invariant class, where the reaching path, feature co-Lipschitz constant, and head slope are all known a priori, so the bracket is genuinely two-sided (not tautological).

Lemma 2 (Radial co-Lipschitz bound for the power spectrum). *Let $\Psi(x) = (|\hat{x}_k|^2)_k$ with the unitary DFT, and for $r_0, \sigma_0 > 0$ let $\mathcal{C}(r_0, \sigma_0, B) = \{x : r_0 \leq \|x\|_2 \leq B, \|\Psi(x)/\|x\|_2\|_2 \geq \sigma_0\}$. For $x \in \mathcal{C}(r_0, \sigma_0, B)$ and $t \in [0, 1]$, $\|\Psi(x) - \Psi(tx)\|_2 \geq r_0\sigma_0\|x - tx\|_2$.*

Proof. Two-homogeneity gives $\Psi(tx) = t^2\Psi(x)$, so $\|\Psi(x) - \Psi(tx)\|_2 = (1 - t^2)\|\Psi(x)\|_2$ while $\|x - tx\|_2 = (1 - t)\|x\|_2$; the ratio is $(1 + t)\|\Psi(x)\|_2/\|x\|_2 \geq \|x\|_2\|\Psi(x)/\|x\|_2\|_2 \geq r_0\sigma_0$ on the cone, using $\Psi(x) = \|x\|_2^2\Psi(x)/\|x\|_2$ and $1 + t \geq 1$. $\square$

Proposition 3 (Certifiable bracket for radial power-spectrum scores). *Let $g = h \circ \Psi$ with Ψ as above, $x \in \mathcal{C}(r_0, \sigma_0, B)$, $g(x) > 0$, $g(0) < 0$; by continuity there is $t_\star \in (0, 1)$ with $g(t_\star x) = 0$, set $z = t_\star x$. If $h(\Psi(x)) - h(\Psi(z)) \geq h_- \|\Psi(x) - \Psi(z)\|_2$ for some $h_- > 0$, then $r_2(x) \leq g(x)/(h_- r_0 \sigma_0)$. If moreover h is h_+ -Lipschitz on $\Psi(B_2(0, B'))$ with B' large enough that the certified neighborhood of x stays in the ball, then*

$$\frac{g(x)}{2B'h_+} \leq r_2(x) \leq \frac{g(x)}{h_- r_0 \sigma_0}, \quad \kappa \leq \frac{2B'h_+}{h_- r_0 \sigma_0}.$$

Proof. $\varphi(t) = g(tx)$ is continuous with $\varphi(0) < 0 < \varphi(1)$, giving $t_\star$ and the boundary point $z = t_\star x$ on the explicit radial path. Lemma 2 and the head secant give $g(x) - g(z) \geq h_- r_0 \sigma_0 \|x - z\|_2$, so $r_2(x) \leq \|x - z\|_2 \leq g(x)/(h_- r_0 \sigma_0)$. For the lower arm, $|g(u) - g(v)| \leq h_+ \|\Psi(u) - \Psi(v)\|_2 \leq 2B'h_+ \|u - v\|_2$ on the ball (Lemma 5), and the standard certificate applies. The reaching path is the explicit radial path, the feature co-Lipschitz constant is the explicit cone constant $r_0 \sigma_0$, and h_- is a property of the head on a known feature segment, so all three are available before any attack is run: the bracket is genuinely two-sided on the cone, not obtained by naming the unknown secant slope of the true minimal path. $\square$

A.4 Result 3: scale-invariance of the ratio (gauge-freeness)

We restate Lemma 3 of the main text, prove it, and record the corollary that the direct maximizer is the constant classifier.

Lemma 3 (Scale-invariance of the ratio; restated). *The per-sample ratio $M(x)/\|\nabla M(x)\|$ is invariant to positive rescaling of the score, $f \mapsto cf$ for $c > 0$, since numerator and denominator scale together.*

Proof. Under $f \mapsto cf$ the margin $M(x) = f_y(x) - \max_{j \neq y} f_j(x)$ scales to $cM(x)$ and $\nabla M(x)$ to $c\nabla M(x)$, so $cM(x)/\|c\nabla M(x)\| = M(x)/\|\nabla M(x)\|$: the ratio, and the input-space radius it lower-bounds, are unchanged. $\square$

Corollary 4 (The direct maximizer is the constant classifier). *An objective that maximizes η/L directly (equivalently penalizes $\|\nabla M\|/M$ over correctly classified points) is blind to the absolute margin and admits the constant classifier $\nabla M \equiv 0$ as a global optimizer, so η/L is ill-posed as a training objective and well-posed only as a diagnostic read at a fixed logit scale.*

Proof. The penalty $\mathbb{E}[\|\nabla M\|/M] \geq 0$ attains its infimum 0 only as $\nabla M \rightarrow 0$, i.e. at a constant score, which classifies at chance. Because the ratio is scale-invariant (Lemma 3) the objective cannot distinguish a large-margin robust classifier from a degenerate flat one; this is the constant-function degeneracy noted by Tsuzuku et al. [27] in the large-target limit, reached here through scale-invariance. $\square$

A.5 Result 4: the nonlinear escape (power-spectrum span and radius)

The invariant *linear* family is one-dimensional (only f_{dc}); nonlinear invariants are richer. We prove the power-spectrum span lemma, the conditional robust-radius corollary, the ball Lipschitz bound, and the exactly-solvable nonlinear invariant model the main text mentions as the “nonlinear escape.”

Lemma 4 (Power-spectrum span). *Define the circular cross-correlation $(w \star x)_s = \sum_j w_j x_{j+s \bmod d}$ and $\Phi_w(x) = \frac{1}{d} \sum_s (w \star x)_s^2$. Then for real w, x , $\Phi_w(x) = \sum_k |\hat{w}_k|^2 |\hat{x}_k|^2$. Varying a real filter w spans the nonredundant real power coordinates $|\hat{x}_0|^2, |\hat{x}_{d/2}|^2$ (when d is even), and the conjugate-pair sums $|\hat{x}_k|^2 + |\hat{x}_{d-k}|^2$ for $0 < k < d/2$; since x is real these exhaust its power spectrum. A complex filter isolates each individual $|\hat{x}_k|^2$. These features are shift-invariant and supported on V_{ac} for $k \neq 0$.*

Proof. The unitary-DFT convolution theorem gives $\widehat{w \star x}_k = \sqrt{d} \widehat{w}_k \hat{x}_k$ up to the correlation sign convention. By Parseval, $\Phi_w(x) = \frac{1}{d} \sum_s |(w \star x)_s|^2 = \frac{1}{d} \sum_k |\widehat{w \star x}_k|^2 = \sum_k |\hat{w}_k|^2 |\hat{x}_k|^2$. For real w , $\hat{w}_{d-k} = \overline{\hat{w}_k}$, so $|\hat{w}_{d-k}|^2 = |\hat{w}_k|^2$: a real filter cannot split a conjugate pair but can isolate it (support $\hat{w}$ on $\{k, d-k\}$ with conjugate values keeping w real), and can isolate the self-conjugate

$k = 0$ and, when d even, $k = d/2$. Linear combinations give the stated real span; a complex filter with a single nonzero coefficient isolates $|\hat{x}_k|^2$. $\square$

Corollary 5 (Conditional robust radius). *For an invariant feature map Ψ with achievable head margin $\eta = \frac{1}{2} \text{dist}(\text{conv } \Psi(X_+), \text{conv } \Psi(X_-))$ and Lipschitz constant L , the induced classifier $g(x) = a^\top \Psi(x) + \beta$ ($\|a\|_2 = 1$, chosen as the hard-margin separator) has $\text{SC} = 1$ and $r_2 \geq \eta/L$.*

Proof. Invariance of Ψ gives $g(R_g x) = g(x)$, so $\text{SC} = 1$. For any data point and admissible δ , $|g(x + \delta) - g(x)| = |a^\top (\Psi(x + \delta) - \Psi(x))| \leq \|a\|_2 \|\Psi(x + \delta) - \Psi(x)\|_2 \leq L \|\delta\|_2$; if $\|\delta\|_2 < \eta/L$ then $|g(x + \delta) - g(x)| < \eta \leq yg(x)$, so the sign cannot change and $r_2 \geq \eta/L$. The content is which Ψ a shift-invariant architecture realizes (Lemma 4); the inequality itself is the standard Lipschitz–margin bound, a lower-bound certificate with no general converse (Remark 1). $\square$

Lemma 5 (Power-spectrum Lipschitz bound on a ball). *Let $Q(x) = (|\hat{x}_k|^2)_k$ with the unitary DFT. On $\|x\|_2 \leq B$, $\|Q(x) - Q(y)\|_2 \leq 2B\|x - y\|_2$, and a scalar quadratic invariant $f_a(x) = \sum_k a_k |\hat{x}_k|^2$ has $\|\nabla f_a(x)\|_2 \leq 2B\|a\|_\infty$.*

Proof. For complex z, z' , $||z|^2 - |z'|^2| \leq |z - z'|(|z| + |z'|)$. Applying this coordinatewise to Fx, Fy and using unitarity of F , $\|Q(x) - Q(y)\|_2 \leq \|F(x - y)\|_2 (\|Fx\|_2 + \|Fy\|_2) = \|x - y\|_2 (\|x\|_2 + \|y\|_2) \leq 2B\|x - y\|_2$. For the gradient, $\nabla f_a(x) = 2F^* \text{diag}(a)Fx$ in the real identification, so $\|\nabla f_a(x)\|_2 \leq 2\|a\|_\infty \|x\|_2 \leq 2B\|a\|_\infty$. $\square$

Proposition 4 (An exactly solvable nonlinear invariant model). *Let U, V be orthogonal subspaces of $\mathbb{R}^d$ and let $X_+ = \{u \in U : \|u\|_2 = a\}$, $X_- = \{v \in V : \|v\|_2 = a\}$ (or finite shift-orbit subsets whose hulls contain 0). The quadratic invariant $f(x) = \|P_U x\|_2^2 - \|P_V x\|_2^2$ classifies X_+, X_- correctly, every point has exact ℓ_2 robust radius $a/\sqrt{2}$, and no linear classifier separates the classes.*

Proof. For $x \in X_+$, $P_V x = 0$ and $\|x\|_2 = a$, so $f(x) = a^2 > 0$; the negative class is symmetric. Decompose a perturbation as $\delta = \delta_U + \delta_V + \delta_W$ with $W = (U \oplus V)^\perp$. The boundary condition $f(x + \delta) \leq 0$ reads $\|x + \delta_U\|_2^2 \leq \|\delta_V\|_2^2$; δ_W cannot help, so set it to 0. With $r = \|x + \delta_U\|_2$, the smallest $\|\delta_V\|_2$ is $|a - r|$ (move along the ray through x) and the smallest feasible $\|\delta_V\|_2$ is r , so $\|\delta\|_2^2 \geq (a - r)^2 + r^2$, minimized at $r = a/2$ with value $a^2/2$; hence the radius is $a/\sqrt{2}$, attained by $\delta_U = -(a/2)(x/a)$, $\|\delta_V\|_2 = a/2$. No affine hyperplane separates the classes because 0 lies in both class convex hulls, so the hulls intersect. This is a point with radius $a/\sqrt{2}$ under a quadratic invariant that no linear classifier matches: the “nonlinear escape.” $\square$

A.6 Result 5: radial–tangential split and the envelope

These back the main text’s claim that η/L is a diagnostic, not a target (§7). We prove the exact polar identity (Proposition 5) and the envelope-via-achieved-margin proposition, keeping the “KKT point need not be globally max-margin” scoping.

Proposition 5 (Radial–tangential split of the margin gradient; restated). *For a bias-free network with degree-one positively homogeneous activations (ReLU, leaky-ReLU, absolute value, maxout), M is degree-one homogeneous. Writing $r = \|x\|_2$, $u = x/r$, $m(u) = M(u)$, the input gradient splits into orthogonal radial and tangential parts $\nabla M(x) = m(u)u + \nabla_{\mathbb{S}} m(u)$ with $u \perp \nabla_{\mathbb{S}} m(u)$, so*

$$\|\nabla M(x)\|_2 = \frac{M(x)}{\|x\|_2} \sqrt{1 + \|\nabla_{\mathbb{S}} \log m(u)\|_2^2}.$$

The radial term is Euler’s identity $\langle \nabla M(x), x \rangle = M(x)$, giving the floor $\|\nabla M(x)\|_2 \geq M(x)/\|x\|_2$ and the cap $\eta/L \leq \sup_i \|x_i\|_2$. The upper envelope $\|\nabla M(x_i)\|_2 \leq \tau M(x_i)/\|x_i\|_2$ holds exactly when $\|\nabla_{\mathbb{S}} \log m(u_i)\|_2 \leq \sqrt{\tau^2 - 1}$; equivalently $\tau_i = \sec \angle(\nabla M(x_i), x_i)$, distinct from the bracket condition number $\kappa = L/\alpha$.

Proof. Degree-one homogeneity gives $M(x) = r m(u)$. For a perturbation $h = au + z$ with $z \perp u$, $Dr(x)[h] = \langle u, h \rangle = a$ and $Du(x)[h] = z/r$ (tangent to the sphere), so $DM(x)[h] = a m(u) + \langle \nabla_{\mathbb{S}} m(u), z \rangle = \langle m(u)u + \nabla_{\mathbb{S}} m(u), h \rangle$ for all h ; hence $\nabla M(x) = m(u)u + \nabla_{\mathbb{S}} m(u)$. Orthogonality gives $\|\nabla M(x)\|_2^2 = m(u)^2 + \|\nabla_{\mathbb{S}} m(u)\|_2^2$, and $m(u) = M(x)/\|x\|_2$ with $\nabla_{\mathbb{S}} \log m = \nabla_{\mathbb{S}} m/m$ gives the displayed form and the envelope equivalence. The radial component equals the Euler form because $\langle \nabla M(x), x \rangle = r \langle \nabla M(x), u \rangle = r m(u) = M(x)$. $\square$

Proposition 6 (Envelope via the achieved margin). *Let $M_\theta(x_i)$ be the signed margin at $x_1, \dots, x_n \in \mathbb{R}^d \setminus \{0\}$, parameter-homogeneous of degree p ($M_{a\theta} = a^p M_\theta$). Let C satisfy $C(a\theta) = a^p C(\theta)$ and $\|\nabla_x M_\theta(x_i)\|_2 \leq B_i C(\theta)$ at differentiability points, and let the limiting gradient-descent direction $\bar{\theta}$ make $x \mapsto M_{\bar{\theta}}(x)$ positively 1-homogeneous with $\min_i M_{\bar{\theta}}(x_i) > 0$. With the achieved normalized margin $\gamma_{\text{IB}} = \min_i M_{\bar{\theta}}(x_i)/C(\bar{\theta})$ and the normalization $\hat{\theta} = (\min_i M_{\bar{\theta}}(x_i))^{-1/p} \bar{\theta}$, at every support point $M_{\hat{\theta}}(x_i) = 1$ and*

$$\|\nabla_x M_{\hat{\theta}}(x_i)\|_2 \leq \frac{B_i}{\gamma_{\text{IB}}}, \quad \|\nabla_{\mathbb{S}} \log m(u_i)\|_2 \leq \sqrt{\left(\frac{B_i \|x_i\|_2}{\gamma_{\text{IB}}}\right)^2 - 1}.$$

Proof. γ_{IB} is scale-invariant: $\bar{\theta} \mapsto a\bar{\theta}$ scales $\min_i M_{\bar{\theta}}(x_i)$ and $C(\bar{\theta})$ both by a^p . With $\alpha = (\min_i M_{\bar{\theta}}(x_i))^{-1/p}$, parameter homogeneity gives $M_{\hat{\theta}}(x_i) = \alpha^p M_{\bar{\theta}}(x_i) = M_{\bar{\theta}}(x_i)/\min_j M_{\bar{\theta}}(x_j)$, so $\min_i M_{\hat{\theta}}(x_i) = 1$, and $C(\hat{\theta}) = \alpha^p C(\bar{\theta}) = 1/\gamma_{\text{IB}}$. The gradient-control hypothesis gives $\|\nabla_x M_{\hat{\theta}}(x_i)\|_2 \leq B_i C(\hat{\theta}) = B_i/\gamma_{\text{IB}}$; dividing by $m(u_i) = M_{\hat{\theta}}(x_i)/\|x_i\|_2$ in Proposition 5 gives the spherical form. $\square$

Remark 3 (Scope of the envelope). For a depth- D bias-free network with 1-Lipschitz 1-homogeneous activations and $C(\theta) = (\|\theta\|_2/\sqrt{D})^D$ the hypothesis holds with $B_i = \sqrt{2}$, since $\nabla_x M = J^\top(e_y - e_{j^*})$ with $\|e_y - e_{j^*}\|_2 = \sqrt{2}$ and $\|J\|_2 \leq \prod_\ell \|W_\ell\|_2 \leq \prod_\ell \|W_\ell\|_F \leq (\|\theta\|_2^2/D)^{D/2} = C(\theta)$ by arithmetic-geometric-mean. Gradient descent on homogeneous networks converges to a KKT point of the max-margin program [18, 14], which for ReLU networks need not be globally max-margin [28]; the bound therefore uses the *achieved* normalized margin γ_{IB} , recovering the global form only when the two coincide. The constant cannot be data-independent: for $(\pm e_1, \pm 1), (\pm e_2/A, \pm 1)$ in $\mathbb{R}^2$ the hard-margin solution is $w^* = (1, A)$, giving $\|\nabla_{\mathbb{S}} \log m(e_1)\|_2 = A \rightarrow \infty$; the linear case is exact with $\tau_i = \|x_i\|_2/\gamma$ [25]. Empirically the factor is large, $\tau \approx 26$ on the trained networks, exactly the slack that lets a gradient penalty lower $\|\nabla M\|$ while the margin falls with it, so η/L diagnoses robustness but is not a target.

A.7 Result 6: the lazy-regime cluster theorem

We prove that orbit-averaging is an orthogonal projection onto the invariant RKHS subspace (non-expansive), so the tied minimum-norm interpolant is never worse than the unconstrained one; on orbit-closed data the two *coincide*, so any empirical advantage is a realized finite-width effect. We keep the smooth-activation caveat (bare ReLU NTK not Lipschitz). We also record the orbit-bundle and orbit-rank lemmas and the degeneracy propositions the main optimization discussion invokes.

Lemma 6 (Orbit-bundle equivalence). *A one-hidden-layer circular convolutional network with global average pooling, $f(x) = \sum_r \alpha_r \frac{1}{d} \sum_{s \in G} \sigma(\langle w_r, S_s x \rangle + b_r) + \beta$, equals a two-layer fully connected network whose hidden units are tied into shift-orbit bundles: $f(x) = \sum_r \sum_s \frac{\alpha_r}{d} \sigma(\langle S_{-s} w_r, x \rangle + b_r) + \beta$, each filter contributing the orbit $\{S_{-s} w_r\}$ sharing output weight α_r/d and bias b_r . The map is exactly invariant.*

Proof. Orthogonality of S_s gives $\langle w_r, S_s x \rangle = \langle S_s^\top w_r, x \rangle = \langle S_{-s} w_r, x \rangle$, which substituted into the pooling expression gives the tied two-layer form. Invariance follows by reindexing the pooling sum: $f(S_t x)$ replaces the sum over s by a sum over $s' = s + t$ ranging over all of G , returning $f(x)$. $\square$

Lemma 7 (Rank of a shift orbit). *For a base pattern $u \in \mathbb{R}^d$, the cyclic orbit $\{S_s u\}_{s \in \mathbb{Z}_d}$ has rank exactly $\#\{k : \hat{u}_k \neq 0\}$, the number of nonzero Fourier coefficients of u .*

Proof. The matrix of orbit columns is circulant, hence diagonalized by the DFT with eigenvalues $\propto \hat{u}_k$; multiplication by the invertible DFT preserves rank, so the rank equals the number of nonzero eigenvalues. $\square$

Proposition 7 (The canonical orbit bases are degenerate, in opposite ways). *Let each class be the shift orbit of a single base pattern. (i) For a single cosine of frequency k , Lemma 7 gives orbit rank 2 (it spans $\{\cos, \sin\}$ at frequency k), violating the near-orthogonal cluster assumption of Frei et al. [9], Li et al. [16], and the signed orbit mean, the direction carrying the non-robustness conclusion there, is **0**. (ii) For a localized spike e_j , the orbit is full-rank and orthonormal (the cluster assumption holds) but the power spectrum is phase-blind, so $\text{Sep}(\Psi) = 0$ and the invariant nonlinear classifier*

cannot separate the two classes. Neither canonical base is a valid testbed for the re-pointing hope: in (i) the non-robustness premise fails, in (ii) the invariant-separation conclusion fails.

Proof. (i) A real cosine of frequency $k \neq 0, d/2$ has $\hat{u}$ supported on $\{k, d - k\}$, so by Lemma 7 the orbit rank is 2; its columns span the two-dimensional $\{\cos, \sin\}$ block, far below the $\Omega(d)$ near-orthogonal directions the cited non-robustness results need. The signed orbit mean $\frac{1}{d} \sum_s S_s u$ projects onto the DC line, which a zero-mean cosine annihilates, giving $\mathbf{0}$. (ii) $\mathcal{O}(e_j)$ is the standard basis (full rank, orthonormal); but $|\hat{e}_{j,k}|^2 = 1/d$ for all k , so $+e_j$ and $-e_j$ (indeed any two spikes) share a power spectrum and no power-spectrum feature separates them, $\text{Sep}(\Psi) = 0$. Numerically for $d = 64, k = 5$ the cosine orbit has rank 2 and signed mean of norm 1.7×10^{-16} , and the spike orbit has rank d and power-spectrum gap 0. $\square$

Proposition 8 (The degeneracy is non-generic). *If a base u has all Fourier coefficients nonzero, its orbit is full-rank (Lemma 7); and two such bases with distinct power spectra are power-spectrum separable, with feature margin $\frac{1}{2} \|Q(u) - Q(v)\|_2$. Hence generic single-orbit data is simultaneously full-rank and invariant-separable, so the degenerate corner of Proposition 7 does not cover single-orbit data in general.*

Proof. Full rank is Lemma 7. The power spectrum is shift-invariant (Lemma 1), so each orbit maps to one point $Q(u)$, $Q(v)$; if $Q(u) \neq Q(v)$ the maximum-margin separator between two points has margin half their Euclidean distance. For u drawn from any law absolutely continuous w.r.t. Lebesgue measure, each event $\hat{u}_k = 0$ is a proper linear constraint of probability zero, and for independent continuous u, v the equalities $|\hat{u}_k|^2 = |\hat{v}_k|^2$ for all k hold with probability zero; hence the generic claim. $\square$

Theorem 2 (Lazy-regime cluster theorem: no minimum-norm certified-Lipschitz gap on orbit-closed data). *Work in the lazy/NTK regime, where each trained network equals the minimum-RKHS-norm interpolant of its fixed neural tangent kernel [13]: write $K(x, x') = \langle \Phi(x), \Phi(x') \rangle_{\mathcal{H}}$ for the tangent kernel with feature map Φ and RKHS $\mathcal{H}_K$, and assume a smooth activation so that Φ is L_Φ -Lipschitz on the data ball $\{\|x\|_2 \leq B\}$. Let the data be orbit-closed with labels constant on orbits, let the kernel be shift-invariant ($K(S_s x, S_s x') = K(x, x')$), and let $\Pi_G f = \frac{1}{d} \sum_s f(S_s \cdot)$ be the orbit-averaging (Reynolds) operator, whose range $\mathcal{H}^G = \Pi_G \mathcal{H}_K$ is the invariant subspace realized by the shift-tied orbit bundle of Lemma 6. For the unit-margin min-norm problem $\min_f \|f\|_K$ s.t. $y_i f(x_i) \geq 1$, let f_{unc} solve over $\mathcal{H}_K$ and f_{inv} over $\mathcal{H}^G$, and set $L_{\text{unc}} = L_\Phi \|f_{\text{unc}}\|_K$, $L_{\text{inv}} = L_\Phi \|f_{\text{inv}}\|_K$. Then*

- (i) Π_G is the orthogonal projection of $\mathcal{H}_K$ onto $\mathcal{H}^G$, hence non-expansive, with $\|f\|_K^2 = \|\Pi_G f\|_K^2 + \|(I - \Pi_G)f\|_K^2$ [7];
- (ii) $\Pi_G f_{\text{unc}}$ is feasible for the invariant problem, so $\|f_{\text{inv}}\|_K \leq \|\Pi_G f_{\text{unc}}\|_K \leq \|f_{\text{unc}}\|_K$;
- (iii) therefore $L_{\text{inv}} \leq L_{\text{unc}}$, and at the common unit margin η the invariant model has the weakly larger certified ratio, $\eta/L_{\text{inv}} \geq \eta/L_{\text{unc}}$. Moreover, because the RKHS norm is strictly convex and the data is orbit-closed, the unconstrained min-norm interpolant is itself invariant, $f_{\text{unc}} = \Pi_G f_{\text{unc}}$, so the orbit-variance $\|(I - \Pi_G)f_{\text{unc}}\|_K^2$ vanishes and $L_{\text{inv}} = L_{\text{unc}}$: on orbit-closed data there is no strict minimum-norm gap. The cluster-geometry advantage seen empirically is a realized-local-Lipschitz / finite-width effect, not a smaller min-norm certificate.

Proof. (i) For a shift-invariant kernel Π_G is self-adjoint and idempotent on $\mathcal{H}_K$ with spectrum in $\{0, 1\}$, hence the orthogonal projection onto its fixed-point set $\mathcal{H}^G$; the displayed identity is the Pythagorean decomposition for that projection [7]. (ii) Orbit-closed data with orbit-constant labels give $y_i f_{\text{unc}}(S_s x_i) \geq 1$ for all s , so $y_i (\Pi_G f_{\text{unc}})(x_i) \geq 1$: $\Pi_G f_{\text{unc}} \in \mathcal{H}^G$ is feasible, whence $\|f_{\text{inv}}\|_K \leq \|\Pi_G f_{\text{unc}}\|_K$, bounded by $\|f_{\text{unc}}\|_K$ via (i). (iii) For any $f \in \mathcal{H}_K$ and x, x' in the data ball, $|f(x) - f(x')| = |\langle f, \Phi(x) - \Phi(x') \rangle_K| \leq \|f\|_K \|\Phi(x) - \Phi(x')\|_K \leq L_\Phi \|f\|_K \|x - x'\|_2$, so f is $L_\Phi \|f\|_K$ -Lipschitz there; applying this to $f_{\text{inv}}, f_{\text{unc}}$ and using (ii) gives $L_{\text{inv}} \leq L_{\text{unc}}$, and dividing the common η gives the ratio comparison. The orthogonality identity of (i) makes the norm gap exactly $\|(I - \Pi_G)f_{\text{unc}}\|_K^2$. On orbit-closed data each $S_s f_{\text{unc}}$ is feasible with $\|S_s f_{\text{unc}}\|_K = \|f_{\text{unc}}\|_K$, so were f_{unc} non-invariant the average $\Pi_G f_{\text{unc}}$ would be feasible of strictly smaller norm by strict convexity, contradicting minimality; hence $f_{\text{unc}} = \Pi_G f_{\text{unc}}$ and $L_{\text{inv}} = L_{\text{unc}}$. $\square$

Remark 4 (Smooth-activation caveat; what remains open). The norm contraction (i)–(ii) is activation-independent, but step (iii) uses smoothness: the bare two-layer ReLU NTK feature map is *not* Lipschitz (its RKHS contains unit-norm functions of unbounded Lipschitz constant, 2), so for ReLU one must take a smooth activation or carry $\sup_{\|x\|_2 \leq B} \|\nabla \Phi(x)\|$ explicitly in place of L_Φ . Theorem 2 thus proves the cluster advantage in the lazy, smooth-activation regime; the synthetic stress test’s tied-versus-unconstrained η/L gap (orbit-variance ≈ 0.3 , ratio up to $4.6\times$ in the lazy regime) is therefore a realized, finite-width effect. The rich (small-init) regime and the bare-ReLU trajectory, where the fixed-kernel picture breaks down [4] and the realized-Lipschitz selection among equal-norm interpolants is the open question, stay open: this is the rich-regime KKT conjecture the main text leaves open, on the same wall reached by Frei et al. [9], Min and Vidal [20], Li et al. [16].

Corollary 6 (Rich two-layer: the max-margin values coincide). *The no-gap conclusion of Theorem 2 needs only that the norm is G -invariant and convex and that Reynolds averaging preserves the orbit-constant margin constraints. For any G -invariant convex norm $\|\cdot\|$, Jensen gives $\|\Pi_G f\| \leq \|f\|$, and on orbit-closed data $\Pi_G f$ is invariant and feasible, so the invariant and unconstrained unit-margin minimum-norm values coincide. This holds for the variation (atomic) norm of an infinite-width two-layer network; since the directional limit of gradient descent on a homogeneous network is a max-margin (KKT) interpolant [18, 14], which for wide two-layer networks is the minimum-variation-norm max-margin classifier [3], weight sharing cannot raise that max-margin. The variation norm is not strictly convex, so an invariant optimum exists but need not be unique; the open rich-regime question is whether the dynamics select an invariant, lower-local-gradient optimum among equal-norm interpolants, not whether a smaller-norm optimum exists.*

B Full experimental tables and protocols

Every number below is traced to a source file, named under its table. Where a number in the vision-paper table could not be reconciled with a JSON it is flagged in the caption and in APPENDIX_PORT_NOTES.md rather than copied. Correlations use Pearson for the CIFAR/ResNet grids and Spearman for the per-image and RobustBench panels, matching the main text.

B.1 Weak-attack table

The “invariance helps” effect is a weak-attack artifact. Table 2 reports the capacity-matched ResNet arms plus the ported TIPS operator, *standard*-trained (no adversarial training) and attacked with FGSM, PGD-10, and AutoAttack at small ℓ_∞ budgets, the protocol of the invariance-helps reports [23, 29]. Under FGSM at $1/255$ the exactly invariant (exact-APS) and shift-augmented arms sit above the standard baseline, reproducing that literature’s ordering for the built-in-invariance arms; adding TIPS’s auxiliary losses restores its FGSM advantage too; under AutoAttack every arm is at or near zero by $\varepsilon = 2/255$, so the ordering ranks models that are all non-robust. FGSM overstates robust accuracy here by more than a factor of 30 relative to PGD-10 at the same budget. The rotation-equivariant architecture of Wang et al. [29], retrained with their code and recipe, reproduces their weak-attack numbers (Wang CascadedGCNN: clean 87.2%, FGSM 28.53% and PGD-40 1.75% at their headline budget 0.03 in their normalized convention) yet retains only 0.1% under threat-matched AutoAttack (pixel-space APGD at 0.015), and 0.0% at 0.03.

B.2 Per-arm adversarial-training robust accuracies

Under adversarial training the exactly invariant (circular / exact-cyclic) arm, whose shift-consistency is 1.0 by construction, is the least robust in every width, while the threat-matched η/L orders robustness. Table 4 gives the CIFAR-10 ℓ_∞ and ℓ_2 per-arm tables and Table 5 the MNIST and Fashion-MNIST ℓ_∞ tables. The four arms share a channel schedule and so an identical parameter count at each width.

B.3 RobustBench anisotropy table

Among already-robust models η/L no longer orders robustness; the gradient’s anisotropy $A = \|\nabla M\|_1 / \|\nabla M\|_2$ does. Table 6 gives the per-model panel for thirty official RobustBench CIFAR-10 ℓ_∞ models: clean accuracy, robust accuracy (APGD-CE $\varepsilon=8/255$, 100 iters), the anisotropy A , its normalized form $A/\sqrt{d}$, and η/L_1 . The correlation summary follows the table.

Table 2: Standard-trained ResNet arms (width 1.0) under weak and strong attacks; ℓ_∞ budgets in /255. AutoAttack on 512 points (binomial 95% half-width ≈ 0.007 near zero). The FGSM ordering at $\varepsilon=1$ reproduces the invariance-helps reading for the built-in-invariance arms (exact-APS and shift-augmented above standard), while the AutoAttack columns show every arm is non-robust. Source: `paper/results/resnet_scale/tips_weakattack_w1.0.json` (standard, blurpool, aps, aug, tips) and `c10stdaux_tips_s0.json` (tips+aux); PGD is 10-step.

arm	clean	consist.	FGSM ₁	FGSM ₂	PGD ₁	AA ₂	AA ₄
standard	0.931	0.965	0.477	0.244	0.263	0.006	0.000
anti-aliased	0.935	0.978	0.472	0.262	0.220	0.002	0.000
exact (APS)	0.926	1.000	0.497	0.277	0.300	0.004	0.000
shift-aug	0.941	0.973	0.532	0.315	0.306	0.004	0.000
TIPS (operator)	0.929	0.974	0.459	0.229	0.261	0.000	0.000
TIPS + aux losses	0.931	0.973	0.493	0.300	0.269	0.004	0.000

Table 3: Wang CascadedGCNN [29], retrained under their code and recipe, weak versus strong attacks. Their protocol (normalized $[-1, 1]$ space, their eps 0.03 = pixel 0.015) is reported in the top block; the pixel-space threat-matched AutoAttack (deterministic APGD-CE+APGD-T) in the bottom. FGSM at their headline budget reproduces their numbers; AutoAttack erases the robustness. Source: `paper/results/b1_wang/cascaded_s0_eval.json` (their_protocol / pixel_space blocks, $n = 1000$).

<i>Their protocol</i> (normalized eps; clean = 87.2)					
budget (norm.)	0.01	0.02	0.03	0.04	0.05
FGSM	42.27	34.47	28.53	24.08	21.41
PGD-40	20.24	5.53	1.75	0.74	0.34
<i>Pixel-space threat-matched</i> (clean = 88.6)					
budget (pixel)	0.01	0.015	0.03		
PGD-40 (ref)	6.4	1.0	0.2		
AutoAttack (APGD)	2.7	0.1	0.0		

Correlation summary (RobustBench, $n=30$). Lower A means more robust: $\text{Spearman}(A, \text{robust}) = -0.79$ ($p = 2.0 \times 10^{-7}$), bootstrap 95% interval $[-0.92, -0.56]$ (5000 resamples). It survives controlling for clean accuracy (partial -0.77) and for η/L_1 itself (partial -0.65). Restricted to the twenty-nine robust models (dropping the single non-robust *Standard*) the ranking is -0.77 (the jackknife drop-of-*Standard* value). The size η/L_1 detects robustness but ranks it only weakly ($\text{Spearman}(\eta/L_1, \text{robust}) = +0.61$). The cross-check against official AutoAttack gives $\text{Spearman}(A, \text{officialAA}) = -0.79$, partial controlling clean accuracy -0.76 .

B.4 Crown-jewel hardened seventeen-tower four-cell table

On zero-shot CLIP over ImageNet-100 the two-by-two crosses modality (image, text) with perturbation type (average-case invariance, worst-case adversarial). Both invariance cells saturate; only the image-adversarial cell S (full targeted AutoAttack) carries cross-encoder spread; the threat-matched η/L_1 predicts it while the consistencies do not. Table 7 gives the per-tower panel and Table 8 the pre-registered correlation table with confidence intervals.

The four-cell saturation is explicit (means over the 17 towers, standard deviations reported as in the main text, sample convention): image-invariance $SC = 0.977 \pm 0.009$ (range $[0.960, 0.990]$), text-invariance $PC = 0.977 \pm 0.007$ (range $[0.962, 0.989]$); the image-adversarial cell $S = 0.385 \pm 0.244$ (range $[0, 0.69]$, $27.6 \times$ the SC spread by the analysis file’s population-sd ratio); the text-adversarial cell $S_{\text{text}} = 0.896 \pm 0.029$ (range $[0.836, 0.945]$, $4.1 \times$ the PC spread).

Within the seventeen-tower panel, per-tower the per-image η/L_1 vs radius Spearman ranges from $+0.52$ (clip_b16_openai, a floor-dominated non-robust tower) up to $+0.97$ (fare4_b32), with the pooled-robust value $+0.89$; this is the hardened panel and is separate from the ten-robust-encoder set of §5 (per-encoder $+0.74$ to $+0.96$). The single text-hardened tower *leaf_L_text* has among the highest paraphrase-consistency and S_{text} (PC 0.9850, S_{text} 0.9234) despite a middling vision η/L_1

Table 4: Capacity-matched CIFAR-10 dissection under PGD adversarial training, per-arm means. ℓ_∞ block: PGD-AT at $\varepsilon=8/255$ (7 steps), evaluated by AutoAttack at $8/255$; η/L_1 is the threat-matched (ℓ_1 dual) ratio. ℓ_2 block: PGD-AT at $\varepsilon=0.5$ (10 steps), AutoAttack at $\ell_2=0.5$; η/L_2 is the matched ratio. Sources: `paper/results/at_cifar_linf_full10k_20260623_185505.json` and `at_cifar_l2_full10k_20260624_111942.json` (full 10k test set, 2 seeds).

arm	w	clean	consist.	robust acc.	η	L_1	η/L_1
<i>CIFAR-10, $\ell_\infty=8/255$ (AutoAttack)</i>							
standard	32	0.712	0.794	0.375	1.48	35.3	1.190
anti-aliased	32	0.724	0.802	0.388	1.56	36.3	1.191
exact cyclic	32	0.569	1.000	0.261	0.80	21.6	1.104
shift-aug	32	0.694	0.867	0.359	1.35	32.3	1.150
standard	64	0.766	0.820	0.408	1.89	46.1	1.133
anti-aliased	64	0.775	0.823	0.419	1.96	46.4	1.154
exact cyclic	64	0.633	1.000	0.294	1.05	28.8	1.081
shift-aug	64	0.750	0.891	0.395	1.72	42.1	1.100
arm	w	clean	consist.	robust acc.	η	L_2	η/L_2
<i>CIFAR-10, $\ell_2=0.5$ (AutoAttack)</i>							
standard	32	0.804	0.829	0.581	2.40	2.45	0.981
anti-aliased	32	0.815	0.832	0.594	2.52	2.48	1.016
exact cyclic	32	0.711	1.000	0.464	1.54	1.79	0.865
shift-aug	32	0.794	0.902	0.570	2.24	2.28	0.984
standard	64	0.846	0.847	0.609	3.16	3.28	0.964
anti-aliased	64	0.854	0.857	0.628	3.23	3.25	0.995
exact cyclic	64	0.769	1.000	0.511	1.95	2.31	0.843
shift-aug	64	0.841	0.924	0.613	2.86	2.88	0.993

Table 5: MNIST and Fashion-MNIST dissection under PGD adversarial training (ℓ_∞), per-arm means. MNIST evaluated at $\ell_\infty=0.3$, Fashion-MNIST at $\ell_\infty=0.1$, both by AutoAttack. The exactly invariant arm has consistency 1.0 and the lowest robust accuracy in each width. Sources: `paper/results/at_mnist_linf_full10k_20260625_043653.json` and `at_fashion_linf_full10k_20260625_160748.json`.

arm	w	clean	consist.	robust acc.
<i>MNIST, $\ell_\infty=0.3$</i>				
standard	32	0.994	0.945	0.942
anti-aliased	32	0.994	0.953	0.942
exact cyclic	32	0.994	1.000	0.926
shift-aug	32	0.994	0.998	0.939
standard	64	0.994	0.946	0.947
anti-aliased	64	0.994	0.954	0.947
exact cyclic	64	0.994	1.000	0.934
shift-aug	64	0.995	0.999	0.952
<i>Fashion-MNIST, $\ell_\infty=0.1$</i>				
standard	32	0.876	0.684	0.776
anti-aliased	32	0.876	0.683	0.776
exact cyclic	32	0.855	1.000	0.724
shift-aug	32	0.862	0.958	0.747
standard	64	0.886	0.680	0.781
anti-aliased	64	0.885	0.690	0.783
exact cyclic	64	0.868	1.000	0.736
shift-aug	64	0.870	0.964	0.760

(0.01174), placing text robustness as a separate axis raised by text training and not by the vision margin.

B.5 MNIST and Fashion-MNIST anisotropy sweep tables

On low-dimensional families a fixed training budget compresses A into a narrow band where it does not rank; an adversarial-training-strength sweep widens it and restores the ranking law. Table 9 gives the twenty-four cell per-dataset correlations at each evaluation budget. Over the sweep A

Table 6: Thirty RobustBench CIFAR-10 ℓ_∞ models, sorted by robust accuracy. A is the mean gradient anisotropy, $\sqrt{d}=55.4$. Source: `llm_transfer/pilots/c1_tower/robustbench_val/results.json` and `FINAL_REPORT.txt` ($n=1000$ eval, 500 correct points attacked). The in-house APGD ranking agrees with official AutoAttack at Spearman +0.96.

model	clean	robust	A	$A/\sqrt{d}$	η/L_1
Standard	0.948	0.000	35.54	0.641	0.00358
Wong2020Fast	0.842	0.451	25.18	0.454	0.04112
Andriushchenko2020Understanding	0.805	0.485	26.43	0.477	0.04299
Ding2020MMA	0.847	0.501	24.00	0.433	0.02538
Cui2020Learnable_34_10	0.885	0.522	21.68	0.391	0.07622
Engstrom2019Robustness	0.887	0.527	21.34	0.385	0.04883
Cui2020Learnable_34_20	0.894	0.549	21.23	0.383	0.06939
Addepalli2021Towards_RN18	0.814	0.554	21.82	0.394	0.05044
Addepalli2022Efficient_RN18	0.866	0.556	24.43	0.441	0.05918
Huang2020Self	0.833	0.561	20.37	0.367	0.07275
Hendrycks2019Using	0.887	0.562	20.97	0.378	0.05281
Zhang2020Attacks	0.846	0.570	21.31	0.384	0.06717
Sehwag2021Proxy_R18	0.852	0.579	21.75	0.392	0.06051
Sehwag2020Hydra	0.901	0.586	21.47	0.387	0.07087
Chen2021LTD_WRN34_10	0.863	0.592	20.29	0.366	0.06534
Gowal2021Improving_R18_ddpm_100m	0.881	0.604	21.36	0.385	0.05745
Addepalli2022Efficient_WRN_34_10	0.885	0.609	21.96	0.396	0.07148
Jia2022LAS-AT_70_16	0.852	0.612	19.50	0.352	0.06579
Sridhar2021Robust	0.901	0.613	20.05	0.362	0.07828
Sridhar2021Robust_34_15	0.874	0.614	18.30	0.330	0.08935
Gowal2020Uncovering_28_10_extra	0.899	0.624	20.09	0.362	0.08079
Rade2021Helper_ddpm	0.882	0.624	20.57	0.371	0.06741
Wu2020Adversarial_extra	0.885	0.627	19.99	0.361	0.07011
Huang2021Exploring_ema	0.914	0.632	19.08	0.344	0.09382
Rebuffi2021Fixing_28_10_cutmix_ddpm	0.881	0.638	21.54	0.389	0.07073
Pang2022Robustness_WRN28_10	0.886	0.640	20.61	0.372	0.06470
Zhang2020Geometry	0.898	0.659	19.88	0.359	0.04861
Pang2022Robustness_WRN70_16	0.891	0.663	18.85	0.340	0.07432
Rebuffi2021Fixing_70_16_cutmix_ddpm	0.884	0.667	18.68	0.337	0.08127
Rebuffi2021Fixing_106_16_cutmix_ddpm	0.886	0.668	18.08	0.326	0.08224

spans [2.38, 16.59] on MNIST and [3.99, 13.17] on Fashion, and is collinear with training strength ($\text{Spearman}(A, \varepsilon_{\text{train}}) \approx -0.85$ MNIST, -0.81 Fashion), so the diverse RobustBench panel remains the cleaner cross-model evidence.

The mismatched- ℓ_2 sign nuance is that strong high- ε ℓ_∞ training over-fits the ℓ_∞ ball and shrinks the ℓ_2 radius, so on MNIST the raw A vs ℓ_2 radius correlation is +0.48 (“wrong” sign relative to the ℓ_∞ story) and collapses to +0.09 once clean accuracy is controlled; on Fashion the mismatched radius keeps the correct sign (-0.74 raw). Every correlation was recomputed three ways with no gradient masking; the effective-dimension deficit $r \approx (\eta/L_1)/(1 + CA)$ this evidence supports remains, as the main text scopes it, a proposed and well-controlled mechanism rather than a network-level theorem.

B.6 The diffusion data-axis, TM-regression interventions, and the certificate-count mechanism

The diffusion data-axis. Intervening on the training *data* instead of the architecture, through diffusion-augmented adversarial training, obeys the same threat-matched law, an independent axis for the diagnostic. Across adversarial-training budgets $\varepsilon \in \{2, 4, 8, 16\}/255$ the raw threat-matched ratio anti-correlates with robust accuracy (-0.95) while the budget-normalized ratio $\eta/(L\varepsilon)$, the certified radius in units of the attack, collapses the sweep onto one monotone curve (Pearson +0.94); within a fixed budget, which is how the dissections compare models, the raw ratio is the right form. At matched η/L_1 the diffusion-data models sit measurably above the real-only models (+0.10 AutoAttack, $t \approx +5$ in the budget sweep), an offset that tracks the train–test robust gap the data removes and vanishes at $\varepsilon = 16/255$ where there is no robust overfitting left to prevent. The mean test-set ratio is therefore not a sufficient statistic across interventions; the per-sample certified fraction is (Proposition 9), which localizes the limitation to the mean summary rather than to the certificate.

Table 7: Zero-shot CLIP, ImageNet-100, seventeen encoders, full targeted AutoAttack at $\varepsilon=4/255$ (0.01569). “rob.” marks image-adversarial robustness (robust=T for the thirteen adversarially robust towers). SC=shift-consistency, PC=paraphrase-consistency, S =image-adversarial accuracy (AutoAttack), S_{text} =worst-of-eleven-paraphrase accuracy. The text-hardened tower *leaf_L_text* (last row, marked $\dagger$) has the highest paraphrase-consistency tier despite a middling vision η/L_1 ; it is counted among the four non-image-robust towers. Source: `llm_transfer/results/c1_tower/crownjewel_hardened.json` ($n_{\text{eval}}=2000$, $n_{\text{attack}}=400$).

tower	rob.	clean	SC	PC	S	S_{text}	η/L_1
clip_b32_laion2b	F	0.843	0.9598	0.9789	0.000	0.9015	0.00069
fare4_b32	T	0.691	0.9628	0.9690	0.393	0.8611	0.01756
tecoa4_b32	T	0.770	0.9658	0.9675	0.575	0.8500	0.02342
fare4_b16	T	0.760	0.9756	0.9623	0.495	0.8361	0.01963
tecoa4_b16	T	0.794	0.9770	0.9724	0.573	0.8741	0.02569
clip_b16_openai	F	0.844	0.9675	0.9808	0.000	0.8986	0.00049
simclip4	T	0.836	0.9790	0.9738	0.540	0.8864	0.01245
simclip2	T	0.877	0.9824	0.9794	0.340	0.9002	0.01124
clip	F	0.900	0.9816	0.9837	0.000	0.9216	0.00031
clip_l14_laion2b	F	0.891	0.9855	0.9853	0.000	0.9321	0.00025
fare4_cnxt	T	0.803	0.9804	0.9726	0.507	0.8773	0.02151
tecoa4_cnxt	T	0.813	0.9737	0.9732	0.620	0.8937	0.02766
fare4	T	0.853	0.9825	0.9756	0.580	0.8927	0.01220
tecoa4	T	0.875	0.9895	0.9820	0.693	0.9143	0.01636
fare2	T	0.885	0.9798	0.9818	0.320	0.9192	0.01130
tecoa2	T	0.915	0.9901	0.9894	0.578	0.9448	0.02207
leaf_L_text $\dagger$	F	0.887	0.9811	0.9850	0.338	0.9234	0.01174

Table 8: Pre-registered correlations on the seventeen-tower panel, with 95% intervals ($n=17$ unless noted). η/L_1 orders the image-adversarial cell but not the text cell (raw anti-correlation is a clean-accuracy artifact, vanishing under the partial); paraphrase-consistency co-moves with S_{text} because both are average-versus-worst over the same narrow paraphrase family. The per-image row pools the thirteen robust towers within-tower ($n=2600$). Source: `llm_transfer/results/c1_tower/crownjewel_hardened_analysis.json`.

relation	Spearman	95% CI
η/L_1 vs S (image-adversarial)	+0.80	[+0.44, +0.95]
partial, controlling clean accuracy	+0.82	[+0.50, +0.95]
η/L_1 vs S_{text} (raw)	−0.54	[−0.87, −0.08]
partial, controlling clean accuracy	+0.006	[−0.55, +0.51]
PC vs S_{text}	+0.98	[+0.89, +1.00]
PC vs S_{text} , partial clean	+0.77	[+0.20, +1.00]
SC vs S	+0.21	[−0.37, +0.67]
PC vs S	−0.30	[−0.75, +0.22]
per-image η/L_1 vs radius (pooled robust, $n=2600$)	+0.89	[+0.88, +0.90]

Four interventions confirm η/L is not a trainable target. The scale-invariance of Lemma 3 makes the ratio objective degenerate and the radial floor of Proposition 5 ties the gradient to the margin; four controlled interventions on a standard backbone settle the empirical question. (1) *Isotropic gradient penalty* on the threat-matched margin gradient $\|\nabla M\|_1$: only the threat-matched ℓ_1 penalty raises η/L (the input-gradient ℓ_1 norm falls from 38 to 22), while the mismatched ℓ_2 and loss-gradient penalties [22, 8] leave it unchanged; but AutoAttack robust accuracy is statistically unchanged for every moderate penalty (paired $p > 0.4$) and falls under the strongest matched penalty (−0.011, $p = 0.026$), with no gradient masking, because penalizing the gradient lowers the margin alongside it (η falls from 1.58 to 1.03). (2) *Ratio maximization* of $\|\nabla M\|_1/M$ drives the classifier to the trivial constant solution (the scale-invariance degeneracy). (3) *Anisotropic hinge penalty* and (4) *directional penalty* at the adversarial point collapse in the same way: clean accuracy falls from 0.728 to 0.38 and 0.29, AutoAttack from 0.390 to 0.258 and 0.201, and the margin from 1.59 to 0.24 and 0.05, every quantity dropping together with no gradient masking. All four collapse the margin.

The certificate-count mechanism. In the locally affine regime the first-order radius $r_1(x) = M(x)/\|\nabla M(x)\|_1$ is an *exact* ℓ_∞ survival test, so robust accuracy is a lower-tail functional of the r_1 distribution.

Table 9: Anisotropy A versus threat-matched ℓ_∞ robust accuracy over 24 cells per dataset (six training budgets $\times$ two widths $\times$ two seeds), with the partial controlling clean accuracy and, for contrast, the mismatched ℓ_2 (DDN) radius. A ranks robustness in the training norm at the informative budgets; the sign is correct only in the training norm (on the mismatched ℓ_2 radius it is right on Fashion but wrong on MNIST, collapsing to null under the clean-accuracy control). Sources: `paper/results/anisosweep_analysis_mnist_linf.json` and `..._fashion_linf.json` (per-cell rows in `at_{mnist,fashion}_linf_anisosweep_*.json`).

dataset / metric	budget	Spearman($A, \cdot$)	partial clean
<i>MNIST</i> ($A \in [2.4, 16.6]$)			
AutoAttack ℓ_∞	$\varepsilon=0.2$	−0.83	−0.75
AutoAttack ℓ_∞	$\varepsilon=0.3$	−0.89	−0.74
ℓ_2 radius (DDN, mismatched)	—	+0.48	+0.09
<i>Fashion-MNIST</i> ($A \in [4.0, 13.2]$)			
AutoAttack ℓ_∞	$\varepsilon=0.1$	−0.61	−0.53
AutoAttack ℓ_∞	$\varepsilon=0.15$	−0.80	−0.67
ℓ_2 radius (DDN, mismatched)	—	−0.74	−0.26

Proposition 9 (Exact first-order certificate count on affine cells). *Fix $\varepsilon > 0$ and suppose M is affine on $B_\infty(x, \varepsilon)$, so $M(x + \delta) = M(x) + \langle g, \delta \rangle$ there with $g = \nabla M(x)$. Then $\inf_{\|\delta\|_\infty \leq \varepsilon} M(x + \delta) = M(x) - \varepsilon \|g\|_1$, so x survives every ℓ_∞ perturbation of radius ε iff $M(x) \geq \varepsilon \|g\|_1$; equivalently (with $r_1 = \infty$ when $g = 0$ and $M > 0$) iff $M(x) \geq 0$ and $r_1(x) \geq \varepsilon$. If the model is affine on $B_\infty(x, \varepsilon)$ for almost every test point, the population robust accuracy equals the certified fraction, $\text{RobAcc}(\varepsilon) = \mathbb{P}_x(M(x) \geq 0, r_1(x) \geq \varepsilon)$.*

Proof. By affinity, $\inf_\delta M(x + \delta) = M(x) + \inf_{\|\delta\|_\infty \leq \varepsilon} \langle g, \delta \rangle$. Hölder gives $\langle g, \delta \rangle \geq -\varepsilon \|g\|_1$, attained by the feasible $\delta_j^* = -\varepsilon \text{sign}(g_j)$; survival on the whole ball is nonnegativity of the infimum, i.e. $M(x) \geq \varepsilon \|g\|_1$, which for $g \neq 0$ reads $r_1(x) \geq \varepsilon$ and for $g = 0$ reduces to $M(x) \geq 0$. Taking probability over the test distribution gives the count. $\square$

Remark 5 (Reading). The forward implication is the margin-over-local-Lipschitz certificate specialized to the ℓ_∞/ℓ_1 dual pair [12, 27]; the content is the converse: on an affine cell the perturbation $-\varepsilon \text{sign}(\nabla M)$ attains the dual norm exactly, so the certificate is an exact survival test, not merely a lower bound. For a multi-class score M is a minimum of pairwise affine margins, so “ M affine on the ball” says the active rival does not switch there; the +0.998 agreement between the certified fraction and AutoAttack indicates this regime holds at the budgets used, and its breakdown at larger budgets or curvier models is the open edge. The budget-normalized mean law is the scale-family shadow of this count, and the data-axis offset is a tail effect: the certified fraction absorbs the dose coefficient ($+0.053 \rightarrow +0.006$).